



GPS DEMO IN 'C'

■ MANUJ ARORA

The global positioning system (GPS) is a satellite-based navigation and positioning system that was designed by the United States Department of Defense to allow a single soldier or group of soldiers to autonomously determine their position to within 10 to 20 metres of truth. It sends and receives radio signals through satellites (refer Fig. 1) located above the earth. A GPS receiver on the earth receives the transmitted signals and provides location and time information to the user. It provides reliable information in all weather conditions, at all times and anywhere in the world.

Fig. 2 shows the GPS communication system. Fig. 3 shows a typical GPS receiver.

GPS, formally known as the NAVSTAR (navigation satellite timing and ranging), was originally developed for the US Military. Because of its navigation capabilities and access to the technology using small, inexpensive equipment, the government made the system available for civilian use. The system is maintained by the United States government and is freely accessible by anyone with the GPS receiver.

GPS technology

The GPS consists of three major segments: control, space and user. Of these, control and space segments are operated by the US Military.

Control segment. The control segment is responsible for constantly monitoring the satellite health, signal integrity and orbital configuration from the ground. It includes a master control system (MCS), ground antennae and monitor stations.

Space segment. The space segment consists of the constellation of GPS

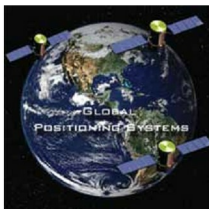


Fig. 1: GPS satellites



Fig. 2: GPS communication system



Fig. 3: Typical GPS receiver

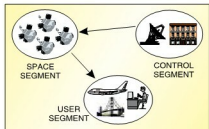


Fig. 4: GPS segments

satellites that are currently in orbit, including operational, backup and inoperable units. The satellites orbit at an altitude of around 20,000 kilometres from ground. A GPS satellite completes orbit once every 11 hours and 58 minutes.

User segment. The user segment consists of the commercially available GPS receiver purchased by the user.

How does a GPS receiver work?

GPS operates on the principle of 'tri-lateralation,' i.e., using information from three known satellites. To find the satellites' positions, it's possible to create imaginary or virtual spheres around each of the satellites.

For the GPS receiver to locate itself, it must find the distance to three satellites of known positions. After detecting a satellite, the receiver assumes that it is located somewhere on the surface of an imaginary sphere centered at the satellite. Let us examine four cases as described below.

When only one satellite is considered. If the GPS receiver calculates only one virtual sphere, the surface of that sphere will intersect the surface of the earth. Because the distance from the satellite to the surface of the sphere is the same all along the circle's circumference, the receiver could theoretically be anywhere on that imaginary sphere (refer Fig. 5).

When two satellites are considered. If the receiver considers two satellites, the second sphere's circle intersects the first sphere's circle and touches the earth at two points. The possible position of the receiver will be in the overlapping region of the two imaginary spheres (refer Fig. 6).

When three or more satellites are considered. A sphere from a third satellite resolves the ambiguity. There